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Integrating machine learning and geo-hydrological modeling to identify rainwater harvesting potential zones in the Upper Tekeze Basin, of Ethiopia

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ABSTRACT

Study region: The Upper Tekeze Basin (UTB), northern Ethiopia, is a semi-arid basin characterized by high rainfall variability, recurrent droughts, rugged topography, and limited surface water availability.
Study focus: This study integrates the Soil and Water Assessment Tool (SWAT), Random Forest (RF) modeling, and geospatial analysis to identify rainwater harvesting (RWH) potential zones. A total of 140 inventory sites (equal suitable/non-suitable) were derived from field surveys, with 70% used for training and 30% for testing. Multicollinearity was low (VIF < 5). SWAT simulated runoff with good performance ($R^2 = 0.77$, NSE = 0.76 for calibration; $R^2 = 0.75$, NSE = 0.75 for validation). RF identified slope (17.5%), drainage density (11.5%), and topographic wetness index (10.1%) as key controlling factors.
New hydrological insights for the region: A fuzzy gamma overlay ($\gamma = 0.9$) with constraints indicated that 16.9% of the basin is suitable for RWH (very high: 2.3%; high: 14.5%). Model performance was strong (AUC = 0.91; accuracy = 81.4%), and field validation confirmed agreement with existing RWH structures. Results highlight the dominant role of topography and runoff processes and demonstrate the effectiveness of integrating hydrological modeling with machine learning for spatial RWH planning.

1. Introduction

Water scarcity is an escalating global challenge, affecting more than 1.2 billion people under physical water scarcity and an additional 1.6 billion facing economic water shortages (United Nations, 2021). The impacts are particularly severe in Sub-Saharan Africa, where livelihoods are heavily dependent on rain-fed agriculture, making communities highly vulnerable to rainfall variability and climate change (Ammar et al., 2016). Increasing climatic variability has intensified the frequency of droughts and dry

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spells, resulting in recurrent crop failures, heightened food insecurity, and growing competition over limited water resources (Al-Hasani et al., 2024).

In Ethiopia, rain-fed agriculture dominates both the national economy and rural livelihoods, employing nearly 85% of the population and contributing about 60% of the national GDP (Abebe and Kassa, 2025). Consequently, the country's socio-economic stability is closely linked to hydro-climatic conditions. The Upper Tekeze Basin (UTB) in northern Ethiopia exemplifies these challenges. The subbasin is characterized by a semi-arid climate, highly variable and uneven rainfall distribution, and rugged topography (Kumie, 2019). These conditions exacerbate surface runoff, soil erosion, and limited groundwater recharge, thereby undermining the reliability of both surface and subsurface water resources (Fentaw et al., 2025). Recurrent droughts and frequent dry spells further constrain agricultural productivity and intensify rural vulnerability, posing significant challenges to sustainable development and water security in the subbasin(Reda et al., 2022).

In response to increasing climate variability, nature-based solutions are increasingly recognized as viable strategies to enhance water availability and resilience in water-stressed regions (Asaye et al., 2025). Rainwater harvesting (RWH) is among the most promising of these interventions and has a long history in Ethiopia. Traditional RWH systems, including ponds, reservoirs, and terracing practices such as those developed by the Konso community, date back to the Axumite period Ethiopia (Alem, 1999; Yemenu et al., 2014). Despite this legacy, the performance of many modern RWH interventions such as ponds and check dams has been inconsistent, often due to poor site selection, seepage losses, and siltation (Fentaw et al., 2025). Nevertheless, RWH remains a cost-effective and environmentally sustainable approach for mitigating seasonal water shortages and improving agricultural productivity, provided that implementation is guided by sound scientific planning (Ahmad et al., 2024; Ghassan et al., 2024)

Conventional RWH planning approaches often fail to adequately integrate key biophysical and socio-economic factors, highlighting the need for rigorous spatial assessments supported by advanced hydrological and decision support tools (Chauhan et al., 2020). In this regard, the integration of Geographic Information Systems (GIS) and Remote Sensing (RS) with Multi-Criteria Decision Analysis (MCDA) methods such as the Analytic Hierarchy Process (AHP) and its fuzzy extension (FAHP) has been widely applied to identify suitable RWH sites, particularly in Ethiopia's semi-arid regions (Abadefar et al., 2025). While these methods provide structured frameworks for ranking suitability criteria, their effectiveness is often constrained by data resolution, subjective expert weighting, and limited representation of upstream-downstream hydrological interactions (AL-Hasani, 2024).

Hydrological models, particularly the Soil and Water Assessment Tool (SWAT), address these limitations by simulating rainfall–runoff processes, thereby improving the hydrological realism of RWH site selection (Abadefar et al., 2025b). Studies conducted in Ethiopia and other African countries have shown that integrating SWAT with GIS-based MCDA significantly enhances the accuracy of runoff estimation leading to more reliable RWH site identification (Umugwaneza, et al., 2022; Abadefar et al., 2025a).

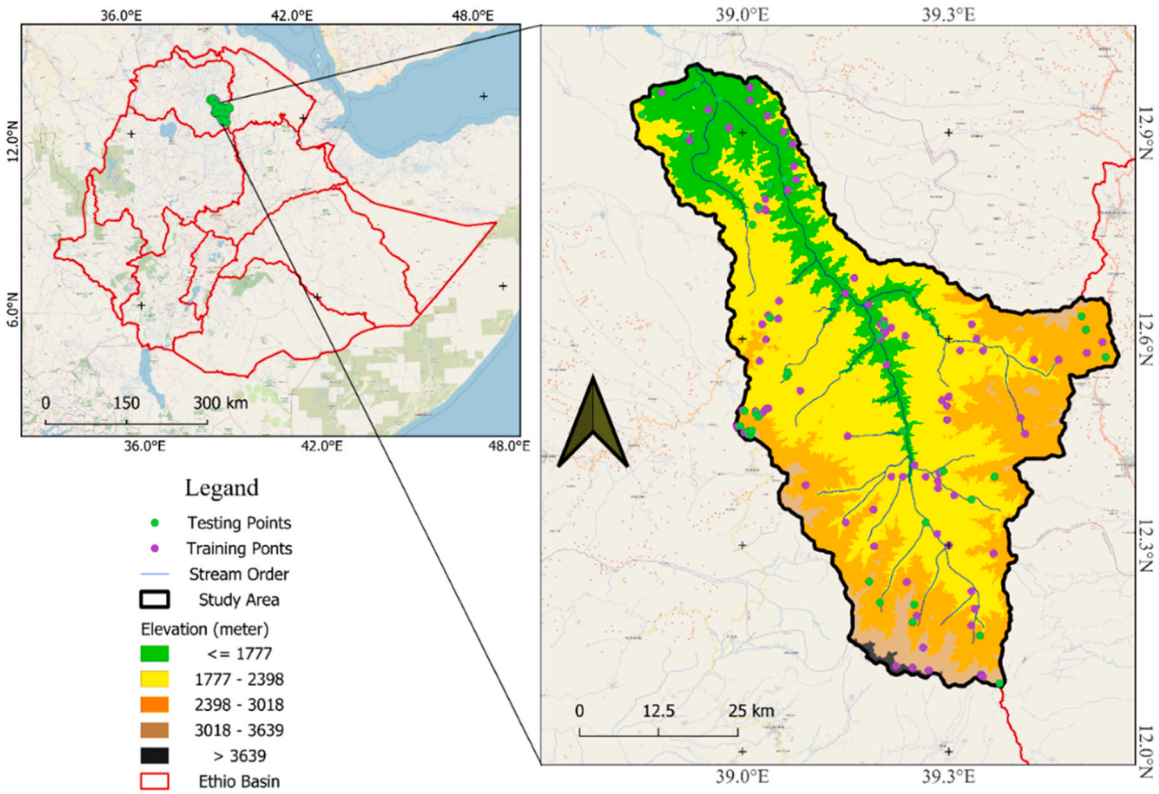


Fig. 1. Location the Upper Tekeze Basin in the northern Ethiopia.

In recent years, machine-learning approaches especially Random Forest (RF) have gained increasing attention in RWH suitability analysis. Traditional MCDA methods such as AHP and FAHP rely heavily on expert judgment and assume linear relationships among criteria, which may limit their predictive performance (Adham et al., 2022). In contrast, RF is a data-driven algorithm capable of capturing complex, non-linear interactions among hydrological, topographic, land-use, and climatic variables, while reducing subjective bias (Chahid et al., 2025; Hasan et al., 2025). Recent studies have demonstrated the superior performance of RF compared to other machine-learning models such as XGBoost and LightGBM in complex, hilly terrains (Hasan et al., 2025).

Integrating RF with GIS/RS and SWAT offers a robust framework that combines spatial detail, hydrological realism, and advanced predictive capability for climate-resilient RWH planning. Building on this integrated approach, the present study aims to identify and validate suitable RWH sites in the Upper Tekeze Basin of northern Ethiopia. Specifically, the objectives are to: (i) estimate surface runoff using the SWAT model; (ii) identify suitable RWH sites based on integrated hydrological, environmental, and spatial criteria; and (iii) validate the proposed RWH suitability mapping framework. The findings are expected to provide valuable insights for policymakers, planners, and local communities to enhance water security and climate resilience in this drought-prone basin.

2. Materials and methods

2.1. Study area

The research was conducted in the Upper Tekeze Basin (UTB) of northern Ethiopia (Fig. 1), covering approximately 3266 km^2 with elevations ranging from 1177 to 4237 m a.s.l. The basin has a predominantly semi-arid climate, receiving a mean annual rainfall of 736.3 mm, about 75% of which occurs during the June to September rainy season (Tela et al., 2024). Rainfall is highly variable, with a prolonged dry season of up to ten months and a short peak rainy period of 50–60 days (Zeleke et al., 2022). The UTB comprises steep uplands and flat to gently sloping lowlands. Dominant soils: Eutric Vertisols, Eutric Cambisols, and Haplic Luvisols cover over 70% of the basin and favor rainwater harvesting due to their moderate to low permeability and high water-holding capacity. Agricultural land (53.92%) and shrubland (25.12%) dominate land use, reflecting high water demand, while bare land (19.25%) contributes high runoff potential. The combination of strong seasonal rainfall concentration, heterogeneous topography, suitable soil properties, and high agricultural water demand makes the UTB a representative and scientifically compelling case for evaluating integrated hydrological and geospatial approaches to RWH site identification in data-scarce, semi-arid highland basins.

2.2. Data sources

Multiple data sets were obtained from various sources and resolution, and utilized (Table 1). Land use/land cover (LULC) was derived from Sentinel-2 imagery (10 m resolution, December 2024), processed using ArcGIS 10.4 and ERDAS Imagine 2015.

A digital elevation model (DEM, 12.5 m) was obtained from the Alaska Satellite Facility. Soil data were collected from the Amhara Design and Supervision Enterprise, while lithology and hydrogeology maps were obtained from the Ministry of Mines. Daily meteorological data from 1990 to 2024 were obtained from the Ethiopian Meteorological Institute. Streamflow data for calibration and validation were obtained from the Ministry of Water and Energy.

2.3. Research methodology

The methodology to identify optimal Rainwater harvesting (RWH) sites integrated Geographic Information Systems/Remote Sensing (GIS/RS), the SWAT hydrological model, and Random Forest (RF) machine learning. Spatial datasets rainfall, Runoff, slope, drainage density, soil texture, lithology, hydrogeology, land use/land cover, topographic wetness index (TWI), terrain ruggedness index (TRI), lineament density, Proximity to urban, road, and well were processed in GIS to generate thematic layers, while remote sensing provided high-resolution land cover information (Ammar et al., 2016; Debebe et al., 2023; Harka et al., 2020).

SWAT simulated rainfall runoff processes and hydrological parameters, including surface runoff, and water balance, to assess potential water availability. Random Forest (RF) was provided non-linear modeling and objective weighting, taking the multiple GIS layers and the SWAT runoff layer and accurately predicting suitability by modeling their complex interactions and GIS for mapping water harvesting (WH) suitability. All factors were prepared as raster with a spatial resolution of 12.5 m. This integrated approach improves reliability, spatial detail, and predictive accuracy, offering a robust framework for sustainable water resource planning in drought-prone regions (Fig. 2).

Table 1
Data sources with their temporal and spatial resolutions.

Data	Source	Time coverage	Spatial Resolution
Sentinel-2 image	Copernicus (www.Copernicus.com)	2024	10 m
DEM	Alaska (https://www.asf.alaska.edu)	2024	12.5 m
Geological data	Ethiopian Ministry of Mines, EMOM	2023	90 m
Soil data	Amhara Design and supervision Enterprise Works (ADSWE)	2022	250 m
Streamflow and well data	Ethiopian Ministry of Water and Energy, EMWE	2003–2018	Point Data
Meteorological data	Ethiopian Meteorological Institute, EMI	1988–2022	Point Data

2.3.1. Field inventory data

The study utilized inventory datasets to delineate RWH zones. A comprehensive field investigation gathered 70 locations categorized as suitable (class value 1) and 70 locations categorized as non-suitable (class value 0) for RWH mapping. These points were randomly split, with 70% assigned to a training set and the remaining 30% to a testing set. This testing set enabled the validation and accuracy assessment of the models employed in the study.

2.3.2. Influencing factors of rainwater harvesting

Rainwater harvesting suitability is influenced by a combination of biophysical and environmental factors which collectively determine runoff generation, storage potential, and site accessibility. Table 2 presents the classification of all influential factors into their respective suitability levels for Rainwater Harvesting (RWH).

2.3.2.1. Soil texture. Soil texture plays a critical role in the effectiveness of RWH site selection as it influences both infiltration rate and water-holding capacity. In this study, soil texture was classified according to its contribution to infiltration and storage potential for RWH identification. Clayey soils, characterized by low permeability and strong water-retention capacity, are therefore highly favorable for water conservation. Fine to medium textured soils are also generally preferred due to their superior moisture and nutrient retention characteristics (Chauhan et al., 2020). Consequently, areas with a higher percentage of clay were assigned higher suitability values (Kumie, 2019), whereas soils with lower clay content received lower suitability ratings due to their relatively high infiltration rates.

2.3.2.2. Ground slope. Ground slope is a critical factor influencing rainwater harvesting (RWH) performance, as it affects runoff velocity, time of concentration, infiltration potential, sediment transport, and groundwater recharge (Adham et al., 2016). Areas with gentle slopes generally promote greater water residence time and infiltration, making them more suitable for RWH implementation, whereas steep slopes tend to generate rapid runoff and increased erosion. Although slopes exceeding 5% are commonly considered less favorable for conventional RWH structures (Critchley et al., 1991), slope suitability may vary depending on local hydrological conditions and the type of harvesting structure. In this study, slope was derived from Digital Elevation Model (DEM) using standard terrain analysis procedures in GIS. The resulting slope values were classified into percentage-based categories following widely applied RWH suitability thresholds reported in the literature. The slope layer was subsequently incorporated as a key spatial criterion in the RWH site suitability analysis.

2.3.2.3. Rainfall. Rainfall is the primary source of all renewable freshwater and a key component of the hydrological cycle, directly influencing groundwater recharge through percolation and surface flow depending on topography (Umar et al., 2016). Missing data

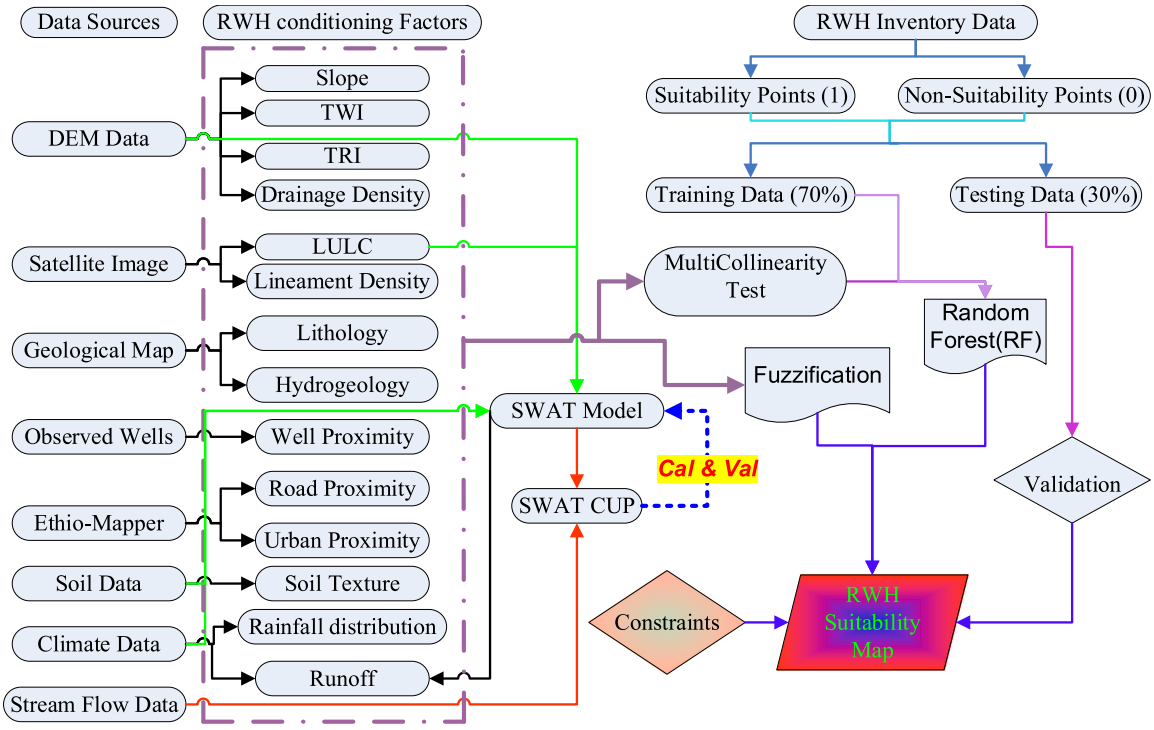


Fig. 2. Framework of the study used to identify suitable RWH sites.

Table 2
RWH Standardized attribute values based on suitability Classes.

Factors	Very Low	Low	Moderate	High	Very High	References
Slope ()	> 30	15–30	8–15	5–8	< 5	(Al-shammari et al., 2021; Senthilkumar et al., 2019)
Soil texture	Lake	Silt clay loam	Loamy sandy	Silt loam	Clay	(Ali, 2017)
Road proximity(km)	< 5	> 20	15–20	10–15	5–10	(Riad et al., 2011)
TWI	< 4.67	4.67– 6.26	6.26 – 8.64	8.64 – 12.4	> 12.4	(Jafarzadeh et al., 2021; Karunanidhi, 2020)
LULU	Built up, Water Body	Forest	Agricultural and shrub land	Grass land	bare land	(Endalkachew et al., 2023; Smakhtin & Aggarwal, 2016)
Rainfall(mm)	< 500	500–750	750–850	850–1000	> 1000	(Kulkarni et al., 2024)
Runoff (mm)	< 50	50 – 100	100–200	200 – 300	> 300	(Debebe et al., 2023)
Drainage Density (km/km^2)	< 2.5	2.5–3.5	3.5 – 4.5	4.5 – 5.5	> 5.5	(Alejandro et al., 2019)
Lineament Density (km/km^2)	> 1.65	1.25–1.65	0.85– 1.25	0.5 – 0.85	< 0.5	(Karunanidhi, 2020)
TRI	> 0.62	0.52 – 0.62	0.43 – 0.52	0.34 – 0.43	< 0.34	(Trivedi et al., 2023)
Hydro geology	lake	-	B5	B4	B3	(Debebe et al., 2023)
Well proximity(km)	< 2.5	2.5 – 5	5–7.5	7.5 – 10	> 10	(Peter et al.; Maha, 2011)
Lithology	Jt, Ja, La	Tv1, Tv2, Tv3, Tv4, Mv/Mb,	MLsst, Talb	Pms, Ttb	Taib, Tab	(Sembroni et al., 2017)
Urban proximity(km)	< 5	> 30	5–10	20–30	10–15	(Debebe et al., 2023)

were infilled using linear regression, short rainfall records were extended accordingly, and point rainfall data were interpolated to generate spatial rainfall surfaces (Abadefar, 2025a). Its magnitude is critical for assessing the suitability of rainwater harvesting (RWH) systems, which require surplus rainfall to be effective (Mechlia et al., 2009). Rainfall exhibits high temporal and spatial variability across a catchment. To estimate rainfall distribution, the Inverse Distance Weighting (IDW) method was applied, which predicts unknown values by averaging nearby observed points (Eq. 1), giving greater weight to closer observations. This approach provides spatially continuous rainfall inputs essential for RWH suitability assessment.

$$Z_g = \sum_{i=1}^{ns} \lambda_i Z_{si} \tag{1}$$

Where Zg is the new estimated value at the required points, Z_{si} is the recorded known value at point i, ns is the total number of observed points and $\lambda = (\lambda_i)$ is the weight contributing to the interpolation.

2.3.2.4. Surface runoff. Surface runoff is a key determinant for rainwater harvesting (RWH) site identification, as it represents the volume of water available for collection following rainfall events. Runoff occurs when rainfall intensity exceeds soil infiltration capacity, and its spatial variability strongly influences the feasibility and effectiveness of surface RWH interventions (Roba et al., 2022).

In this study, spatially distributed surface runoff was simulated using the Soil and Water Assessment Tool (SWAT), a physically based, semi-distributed hydrological model widely applied in semi-arid regions. The model was implemented for the UTB using land use, soil, topographic, and daily meteorological data. Watershed delineation and Hydrologic Response Unit (HRU) definition were performed using threshold values of 10% for land use, 20% for soil, and 10% for slope Asmamaw (2013), resulting in 29 sub-basins and 304 HRUs. Simulations were conducted at a monthly time step for the period 1988–2022, with a three-year warm-up period.

Model calibration and validation were carried out using observed streamflow data and the SUFI-2 algorithm within the SWAT-CUP framework, and model performance was evaluated using NSE, R^2, and PBIAS (Abbaspour et al., 2007). The calibrated SWAT model was then used to generate spatially distributed surface runoff depth, which served as a primary input for the surface RWH suitability analysis.

2.3.2.5. Drainage density. Drainage, defined as the network through which surface water flows to an outlet, is a key factor for RWH site selection (Horton, 1945). A DEM was used to extract the drainage density map, calculated as the total stream length divided by the watershed area (Horton, 1945). High drainage density generally indicates high runoff, while lower-order streams with high infiltration and permeability are more suitable for RWH (Tumbo et al., 2014). These lower-order streams often have less channelization, promoting groundwater recharge. Drainage patterns thus reflect both surface and subsurface hydrological conditions critical for identifying optimal RWH sites. Drainage density was defined by (Horton, 1945) as the ratio of the total length of streams in a watershed over its contributing area as stated in eq.2.

$$D_d = \frac{\sum L_k}{A_k} \tag{2}$$

Where Dd = Drainage density, $\sum L_k$= Total length of all stream segments in a given watershed and A_k = Total area of the watershed.

2.3.2.6. Distance to the road. Proximity to road network is one of the greatest socio-economic criteria for the RWH site selection. Evidence indicates that areas within 100 m of major roads are unsuitable due to elevated risks of water contamination from runoff, pollutants, and sediment deposition. Sites located 100 m to 5 km from roads also tend to receive lower suitability scores, as vehicle emissions and settlement encroachment may adversely affect water quality and long-term performance. Conversely, distances of 5–15 km are generally considered optimal, providing an effective balance between accessibility for construction and maintenance, transportation efficiency, and reduced contamination potential under Ethiopian environmental and socio-economic conditions (Gebremedhn et al., 2023). Beyond approximately 15 km, suitability decreases due to increasing transportation and water-hauling costs, which have been shown to rise by 20–50% per km in rural contexts. These thresholds collectively justify the selection of a moderate distance buffer for RWH siting, enhancing overall feasibility, sustainability, and hydrological performance of storage systems.

2.3.2.7. Distance to urban. The Urban areas are characterized by dense human settlement, extensive built-up infrastructure, and significant economic activity, all of which contribute to elevated risks of water contamination. As a result, locations within 10 km of the urban center were classified as unsuitable for RWH due to pollution sources such as industrial effluents, sewage discharge, and runoff from impervious surfaces. Sites located between 10 and 30 km from the urban center were identified as the most suitable zone, as this distance range provides a balance between water quality protection and economic viability (Girma and Shuniye, 2024). It minimizes implementation and transportation costs, supports gravity-fed or low-energy pumping systems, and reduces the potential exposure of urban populations to waterborne diseases. Beyond 30 km, suitability decreases because logistics, fuel, and transportation demand significantly increase overall project costs unless aided by favorable topography. This classification supports a strategic distance-based buffer for sustainable RWH site selection a finding supported by Tsiko and Haile (2011).

2.3.2.8. Lithology. The lithological units in the study area were reclassified based on their permeability, fracture, and nature of lithological structure (Sembroni et al., 2017). Lithology is the main influence of infiltration. Therefore, permeable and fracture rock formation more suitable for recharge ground water and impermeable bedrock characteristics are more suitable for surface RWH sites (Sembroni et al., 2017).

2.3.2.9. Lineament density. Lineaments were extracted from Sentinel 2 A, a combination of Band B_2 (blue), B_{11} (SWIR-1), and B_{12} (SWIR-2) images, and analyzed using Geomatical 2015 software. Afterthought, lineaments were manually digitized (Chen et al., 2022). The lineaments can be defined as the area of a weak surface that shows some linear to curvilinear features in the geological structure, such as fracture, joint, fault, etc. (Mallick et al., 2019). The extracted lineament was buffered since it represents the spatial influence that would lead to leakage or seepage for the surface water storage (Maina and Raude, 2016). The Lineament Density was calculated using Eq. 3:

$$L_d = \sum_{i=1}^{n} \left(\frac{L_i}{A} \right) \tag{3}$$

2.3.2.10. Land-use/land-cover. Sentinel-2A imagery (6 December 2024) obtained from the Copernicus platform was preprocessed in ArcGIS 10.4 and ERDAS Imagine 2015. Bands B3, B4, and B8 were merged to produce a mosaic of the study area. Ground control points collected by GPS, Google Earth interpretation, previous studies, and the 2020 ESRI 10 m LULC map supported image classification. A supervised maximum likelihood classifier was applied to generate the LULC map. Random sampling points were created for accuracy assessment, resulting in 494 field-verified locations. Classification accuracy was evaluated using a confusion matrix, including overall accuracy, user's accuracy, and the Kappa coefficient, with higher Kappa values indicating strong agreement between reference and classified data (Daniel et al., 2020). The Kappa statistic was calculated using Eq. 4:

$$kappa\ coefficient = \frac{P_o - P_e}{1 - P_e} \tag{4}$$

2.3.2.11. The Terrain Ruggedness Index (TRI). The Terrain Ruggedness Index (TRI) was used to assess terrain heterogeneity and its influence on surface runoff and storage potential for rainwater harvesting (RWH). TRI quantifies local topographic irregularity by calculating elevation differences between a central DEM cell and its eight neighboring cells (Riley, 1999). High TRI values indicate rugged terrain with steep slopes that accelerate runoff, reduce water residence time, and increase erosion risk, making such areas less suitable for surface RWH. Conversely, low TRI values represent level or gently undulating terrain that favors runoff attenuation and surface ponding, supporting structures such as ponds, and check dams (Ganie et al., 2020). TRI was computed from the DEM, normalized using min–max scaling, and integrated with other conditioning factors in the GIS-based suitability analysis. Feature scaling based on minimum–maximum normalization was applied as follows in Eq.5:

$$TRI = \frac{FS\ mean - FS\ \min}{FS\max - FS\ \min} \tag{5}$$

where FS_{mean} denotes the mean TRI value of a grid cell, and FS_{max} and FS_{min} represent the maximum and minimum TRI values across the study area.

2.3.2.12. Topographic Wetness Index (TWI). The Topographic Wetness Index (TWI), or Compound Topographic Index, quantifies the spatial tendency of an area to accumulate water. It integrates the upslope contributing area (α) and local slope (β) using Eq. 6: A higher TWI values occur where large contributing areas coincide with gentle slopes, indicating zones with greater moisture accumulation and prolonged water residence (Kopecký et al., 2021). Such areas are highly suitable for surface RWH structures ponds, check dams, and small reservoirs as they naturally retain runoff and support enhanced infiltration. Conversely, low TWI values correspond to limited contributing areas and steep slopes, where rapid drainage reduces storage potential, making them less appropriate for surface storage, however, sometimes suitable for in-situ techniques such as terracing (Kopecký et al., 2021)

$$TWI = In(\alpha/\tan\beta) \tag{6}$$

2.3.2.13. Hydrogeology. Hydrogeological characteristics of the area were considered to assess the ability of aquifer systems to receive and transmit recharge from RWH structures. The area contains three main hydrogeological units: B3, extensive and highly productive fissured aquifers (5–25 L/s); B4, extensive and moderately productive fissured aquifers (0.51–5 L/s); and B5, extensive low-productivity basement aquifers (<0.5 L/s)(Mengistu et al., 2019). For the suitability assessment, these classes were reclassified according to their recharge potential, with B3 assigned very high suitability, B4 high suitability, and B5 moderate suitability.

2.3.2.14. Well proximity. Proximity to existing groundwater wells was used as a hydrological criterion in the RWH suitability assessment. Euclidean distance was computed in Geographic Information System (GIS) platform to determine the distance of each location from the nearest well (Riad et al., 2011). Areas close to groundwater wells were considered less suitable due to risks of over-abstraction and interference with recharge zones, whereas sites located at moderate distances support sustainable recharge without inducing drawdown conflicts. The resulting distance raster (km) was reclassified into suitability classes, with greater distances assigned higher suitability and near-well areas assigned lower suitability.

2.4. Random forest based criterion weighting

Random Forest (RF) was employed to derive objective criterion weights for the RWH suitability model. The workflow followed a sequential approach (Pappaka et al., 2025). First, the RF classifier was trained using geospatial predictors representing key environmental factors. The model was implemented in Python using scikit learn, and hyperparameters were optimized via cross-validation. The final configuration (900 trees and 70 terminal nodes) was selected based on the minimum out-of-bag (OOB) root mean square error (RMSE).

After tuning, RF aggregated predictions from bootstrapped decision trees and random feature subsets, capturing nonlinear relationships, interactions, and collinearity typical of watershed-scale datasets. Variable importance was quantified using permutation-based mean decrease in accuracy (MDA), which measures the reduction in model performance when each criterion's values are randomly permuted. To retain only statistically meaningful predictors, observed importance scores were compared against null (absence) importance distributions generated from randomized noise features. Criteria exceeding the 95th-percentile null threshold were considered significant. Finally, normalized importance scores were assigned as objective criterion weights for weighted overlay based RWH suitability mapping.

2.5. Information gain-based variable importance

2.5.1. Importance of variables

The Information Gain (IG) is widely used to measure variable importance in machine-learning models because it quantifies the reduction in uncertainty achieved when a predictor is included in the model (Abdar et al., 2021). In this study, IG was applied to identify the most influential attributes for predicting water-resource conditions. Variables with higher IG values were prioritized, enabling the removal of less informative predictors and thereby improving model accuracy and computational efficiency (Zaresefat and Derakhshani, 2023). The calculation of IG is presented in Eq. (7).

$$\mathrm{IG}(\mathrm{T},\mathrm{RF}) = \frac{\mathrm{Entropy}(\mathrm{T}) * \mathrm{Entropy}(\mathrm{T},\mathrm{RF})}{Split\ Entropy(T,RF)} \tag{7}$$

where T is the training dataset, RF is the associated factor, and (T, RF) represents the information gained by dividing the data on the basis of the RWH potential factor.

2.5.2. Multicollinearity detection

Multicollinearity analysis evaluates the degree of correlation among explanatory variables in a machine-learning model, as high correlations can cause significant issues (Al-Hasani et al., 2024). Excessive multicollinearity complicates coefficient estimation, obscuring the individual effects of variables and producing unreliable results (Al-Hasani et al., 2024). To detect multicollinearity, this study employs the variance inflation factor (VIF) and correlation matrix. The VIF is calculated as shown in Eq. (8), with values of VIF > 10 or tolerance < 0.1 indicating potential multicollinearity (O'Brien, 2007).

$$VIF = \frac{1}{Tolerance} \tag{8}$$

2.6. Fuzzy weighted linear aggregation

The appropriate site for RWH was specified using GIS and remote sensing with the SWAT Model integration of the RF approach in the UTB by considering fourteen thematic layers. The five comparable classes are used as indicators for suitable areas for water harvesting suitability: very high suitable, high suitable, moderately suitable, low suitable, and very low suitable. The final stage is to apply the gamma fuzzy overlay to overlay each criteria layer. The Fuzzy Overlay weighted method was used for fuzzy aggregation. This fuzzy overlay was employed to calculate the sum of the weighted criteria. Sadeghi et al. (2019) concluded that, the best operator was gamma fuzzy overlay techniques, with a power of 0.9. The suitability map for RWH was generated by integrating the result of fuzzified raster maps of the layers, each thematic map weighted using the RF model and calculated by map algebra raster calculator were combined through gamma fuzzy overlay techniques to yield the RWH suitability map. Weighting was given by multiplying the weight of the criteria layers by its standardized membership value as shown in Eq. 9.

$$\mu_{Combination} = (FuzzyAlgebraic\ Sum)^{\gamma} \quad * \quad (FuzzyAlgebraicProduc)^{1-\gamma} \tag{9}$$

Where $\mu_{Combination}$ is the suitable site, γ is a parameter between 0 and 1 (Zimmermann, 1980). When γ is 1, the combination is the same as the fuzzy algebraic Sum and when γ is 0, the combination equals the fuzzy algebraic Product. This operator was applied with 0.9 powers.

2.7. Constraints

To ensure constraint, the suitability rating for each assessment criteria was falling within one of the following; either very high, high, moderate, low or very low. A poor rating represents a development constraint that prevents a site from being suitable for RWH. However, in the study major roads, Lake, lineaments, and settlements were considered as some constraints; these criteria have very low water hold capacity (lineament), and the major roads, Lake, Well and settlements places are preferable for socio-economic benefits to RWH sites as shown in the Table 3.

After the constraints were considered in the suitability analysis; the final potential RWH suitability map was generated using the Eq. 10.

$$S = \left((FuzzyAlgebraic\ Sum)^{\gamma} * (FuzzyAlgebraicProduc)^{1-\gamma}\right) * \prod C_i \tag{10}$$

Where, C_i is the criteria score of the constraint i, $\prod$ is the product.

2.8. Criteria for proposing sites for rainwater harvesting structure

In this study, micro-dams and ponds were selected as representative RWH structures suitable for semi-arid highland conditions of the Upper Tekeze Basin (UTB). Site selection and structure type assignment were guided by established hydrological, geomorphological, and land-use criteria reported in the literature (Debebe et al., 2023; Harka et al., 2020).

2.9. Validation

In the development of an RWH potential zone map, the evaluation of predictive accuracy is a crucial step. The area under the receiver operating characteristic (ROC), area under curve (AUC) is a highly effective method for assessing the reliability and quality of susceptibility maps. This widely adopted approach measures the performance of predictive models, providing a robust indication of their predictive capabilities (Chuvieco et al., 2010; Yesilnacar and Topal, 2005). The interpretation of AUC values is as follows: an AUC equal to or below 0.5 indicates no improvement in prediction accuracy over random chance. An AUC in the range of 0.7–0.9 signifies a reasonably good agreement between the model and the observed data. AUC values equal to or greater than 0.9 suggest an optimal scenario where the model's predictions closely align with the actual potential zone of RWH.

Table 3
Constraint factors and the associated justification.

Factors	Limitation (km)	Value	Suitability
Distance to Road	≤ 5	0	Non-Suitable
	> 5	1	Suitable
Lake	≤ 2	0	Non-Suitable
	> 2	1	Suitable
Distance to Lineament	≤ 1	0	Non-Suitable
	> 1	1	Suitable

Source: (Debebe et al., 2023)

Table 4
Linkage between RWH techniques, and site conditions for RWH structure identification.

RWH Technique	Rainfall (mm/year)	Slope (%)	Soil Texture	LULC Type	Porosity & Percolation	Runoff Potential	Stream Order	Suitability Class
Farm Ponds	> 200	< 5	Sandy clay loam, silt loam	Cultivated land, shrubland, grassland	Moderate to low	Medium to high	2nd–3rd order	High to Moderate
Check Dams / Micro Dams	< 1000	< 15	Sandy clay loam, loam	Shrubland, barren land, scrubland	Low	High	4th–6th order	Very High to High

Source: own elaboration based on:(on: (Demisachew et al., 2022;, and Debebe et al., 2023).

3. Results

3.1. LULC classification map

The LULC classification for the UTB demonstrated high accuracy, with overall accuracy ranging from 87.9% to 89% and a Kappa coefficient of 0.89, indicating strong agreement between the classified and reference data (Table 5). These results confirm that the LULC dataset is reliable for subsequent hydrological and rainwater harvesting (RWH) suitability analyses (Shiferaw et al., 2019).

The land use/land cover (LULC) analysis reveals that agricultural land is the dominant class in the basin, accounting for 1764.5 km^2 (53.9%), followed by shrubland (821.9 km^2, 19.3%) and bare land (630.1 km^2, 19.3%). Forest cover is limited to 30.96 km^2 (1.13%), while grassland (0.85 km^2, 0.03%), built-up areas (2.22 km^2, 0.06%), water bodies (1.62 km^2, 0.05%), and lakes (14.31 km^2, 0.43%) occupy negligible proportions (Fig. 3). In terms of rainwater harvesting (RWH) potential, agricultural and bare lands are highly suitable due to their capacity to generate surface runoff and accommodate both in-situ and ex-situ RWH structures (Debebe et al., 2023). Shrubland exhibits moderate suitability as vegetation enhances infiltration, whereas forested areas are less favorable for surface RWH but play a critical role in groundwater recharge. Built-up areas and existing water bodies were excluded from RWH suitability analysis owing to land-use restrictions and pre-existing water storage.

3.2. SWAT mode hydrologic characterization

Spatially distributed surface runoff, generated using the calibrated SWAT model, captured both seasonal and spatial variations across the subbasin (Adeogun et al., 2014). Runoff depths were highest in areas with moderate slopes and less permeable soils, whereas steep or highly permeable zones produced lower runoff volumes. This runoff distribution provides the primary hydrological input for identifying suitable RWH sites.

Parameter sensitivity analysis identified 20 hydrologically influential parameters controlling surface runoff, with the most critical including the Soil Conservation Service Curve Number (SCS-CN2), soil saturated hydraulic conductivity (SOL_K), soil available water capacity (SOL_AWC), baseflow recession constant (ALPHA_BF), groundwater delay time (GW_DELAY), and soil evaporation compensation factor (ESCO) (Table 6). SCS-CN2 controls rainfall–runoff partitioning and directly influences surface runoff volumes, while SOL_K and SOL_AWC regulate infiltration, soil moisture storage, and percolation, affecting both runoff and recharge (Arnold et al., 2012). ALPHA_BF and GW_DELAY determine the timing and magnitude of baseflow contributions, reflecting groundwater–surface water interactions. ESCO influences soil evaporation losses and near-surface moisture dynamics, indirectly modifying

Table 5
Confusion matrix of classification accuracies.

		Confusion matrix of classification accuracies for Upper Tekeze Basin								
		Reference Data								
	Classified data	Bare land	Forest	Grassland	Build Up	Shrub	water	Agriculture	Total	User accuracy (%)
Classified	Bare land	**207**	0	1	3	1	0	6	218	94.9
	Forest	0	**6**	0	0	0	0	0	6	100
	Grassland	0	0	**3**	0	0	0	0	3	100
	Build Up	0	0	0	**7**	0	1	0	8	87.5
	shrub	1	3	1	0	**61**	0	14	80	76.2
	water	1	0	0	0	0	**2**	0	3	66.6
	Agriculture	12	0	0	2	2	0	**561**	577	97.2
	Total	221	9	5	12	64	3	581	**847**	
	Producer accuracy (%)	93.67	66.6	60.00	58.33	95	66.7	96.56	**895**	

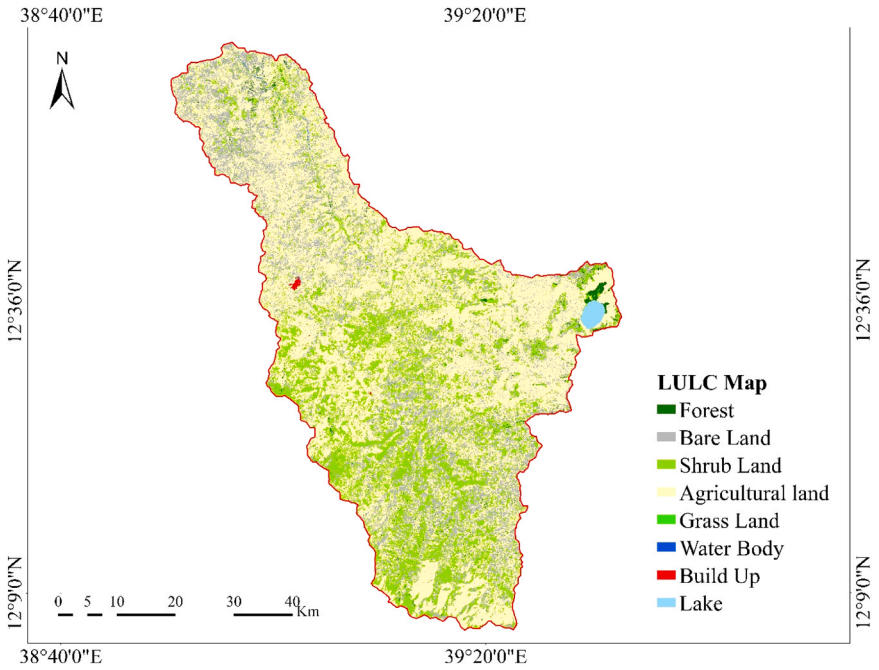


Fig. 3. LULC.

Table 6
Ranking and calibrated values of selected parameters.

Rank	Parameters and Method	Description	T-test	P-test	Initial range	Fitted Value	Final range
1	CN2.mgt(r)	SCS runoff curve number for moisture condition II	-61.1	0.01	74–87	0.021	75.6–88.8
2	SOL-AWC.sol(r)	available water capacity of the soil layer (mm H2O per mm of soil)	43.64	0.014	0.12	0.134	0.134
3	ESCO.hru(v)	Soil evaporation compensation factor	33.81	0.019	0.95	0.56	0.56
4	Surlag.bsn(v)	surface runoff lag coefficient	29.42	0.021	9.25	7.56	7.56
5	OV_N.hru(v)	Manning's "n" value for overland flow	17.17	0.036	0.15	0.348	0.348
6	ALPHA_BF.gw(v)	Baseflow alpha-factor (days)	17.5	0.037	0.048	0.006	0.006
7	GW_REVAP.gw(v)	Groundwater "revap" coefficient	-16.9	0.038	0.02	0.026	0.026
8	GWQMN.gw(v)	threshold depth of water in the shallow aquifer for return flow to occur (mm)	-16.39	0.040	1000	640.5	640.5
9	SOL K. sol(r)	Saturated hydraulic conductivity ($mm \bullet h^{-1}$)	14.99	0.042	0.71	0.405	0.405

A "v" in the method implies a replacement of the initial parameter value with the given value in the final range, whereas an "r "indicates a relative change to the initial parameter value.

runoff generation. Sensitivity to these parameters indicates that accurate representation of soil and groundwater processes is critical for reliable runoff simulation and uncertainty reduction, particularly for RWH and recharge assessments in semi-arid basins (Das et al., 2024). Calibration focused on these sensitive parameters improved model performance and reduced uncertainty in runoff estimation (Zhang et al., 2009).Together, the high-quality spatial datasets and robust runoff simulation establish a strong foundation for RWH site identification, ensuring that both spatial and hydrological variability are accurately represented in the suitability assessment.

Model performance statistics indicated strong agreement between observed and simulated flows (Almeida et al., 2018). During calibration and validation period, the model attained reasonably good performance (i.e., Calibration: $R^2 = 0.77$ and NSE = 0.76; and Validation: $R^2 = 0.75$ and NSE = 0.75, which fall within the "good" to "very good" range according to commonly applied hydrological standards (Moriasi et al., 2015).

Observed and simulated runoff distributions were approximately uniform, with a non-significant difference in mean monthly flows at the 99% confidence level ($t = 1.65 < 1.97$). The model slightly overestimated streamflow, with an overall percent bias (PBIAS) of 5.4%. The results demonstrated that the SWAT model reliably reproduces the hydrological response of the basin and can be confidently used for runoff estimation and hydrological assessments (Fig. 4).

Comparable SWAT applications in the Tekeze River Basin and other Ethiopian highlands report similar model performance. For example, Getu (2015) obtained very good model performance (i.e., $R^2 = 0.82$ and NSE = 0.81 for calibration, and $R^2 = 0.70$ and NSE = 0.63 for validation of monthly stream flow in the Tekeze Basin. Similarly, other regional studies also report monthly NSE and R^2 values

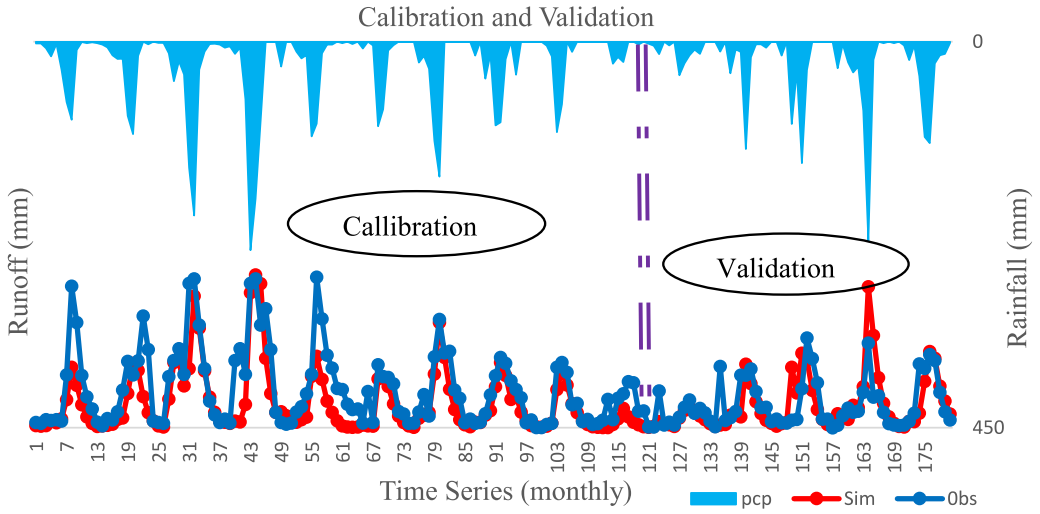


Fig. 4. SWAT model calibration and validation for streamflow at the monthly time scale.

generally between 0.65 and 0.85, with acceptable PBIAS values, indicating satisfactory-to-good model fits(Teshome et al., 2020). The performance metrics obtained in this study are therefore consistent with previous studies and can reasonably applied, for runoff simulation in the Tsitsika watershed.

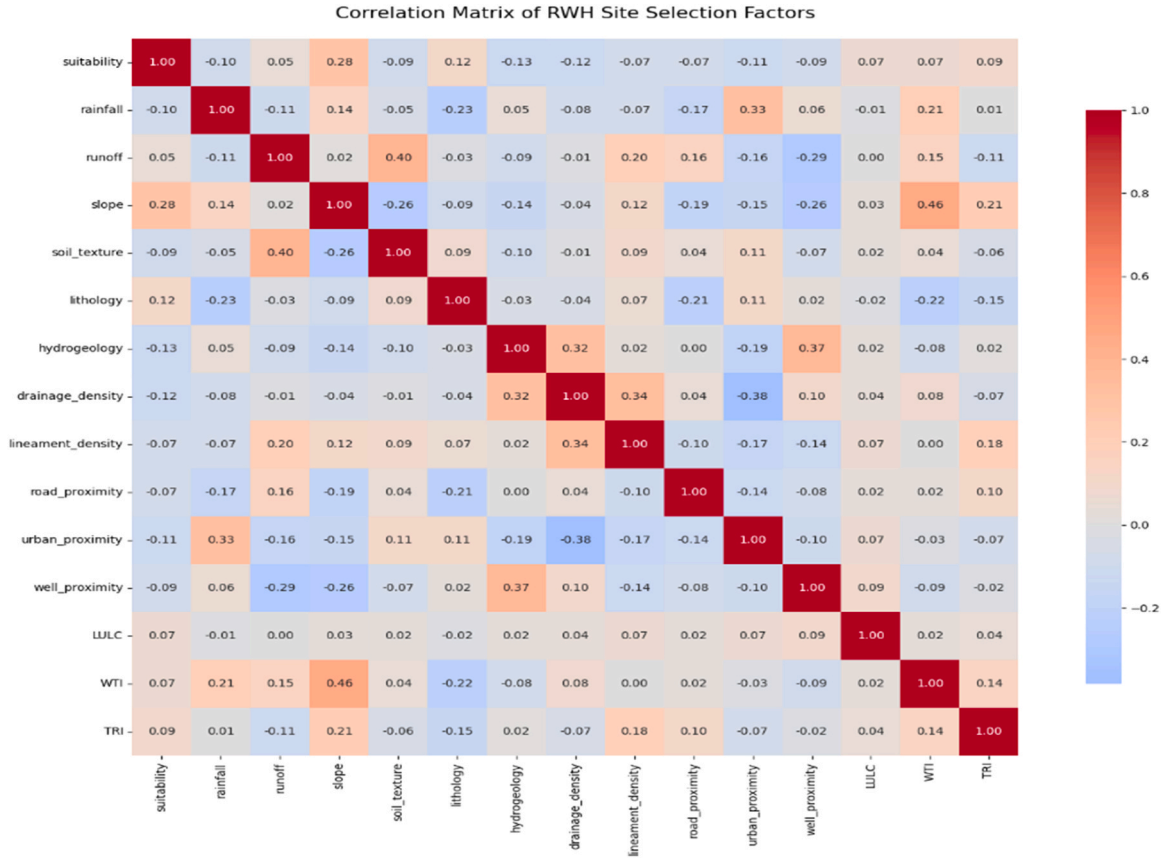


Fig. 5. The correlation matrix showing the magnitude of the relationship among factors.

3.3. Correlation matrix and multicollinearity test

Pappaka et al. (2025) noted that correlations above 0.7 typically depicts collinearity concerns. In this study, the Pearson correlation matrix (Fig. 5) was used to assess linear relationships among RWH site selection factors and the final suitability index. Slope negatively correlated with RWH suitability (r = -0.28), favoring gentle to moderate slopes due to lower runoff, reduced erosion, and easier construction. Similarly, Lithology showed a negative influence (r = -0.23), reflects the dominance of permeable and fractured geological formations in the study area, which promote infiltration and subsurface percolation rather than surface runoff accumulation. Rainfall and runoff were highly correlated (r = 0.94), reflecting the expected hydrological relationship, and justifying retention of runoff as a more comprehensive indicator that incorporates slope and soil effects. Topographic indices were largely independent, with weak WTI to TRI correlation (r = 0.14) confirming that they capture complementary terrain characteristics.

Groundwater well proximity moderately correlated with hydrogeology (r = 0.37), indicating that productive aquifers tend to host more wells, while remaining independent of road and urban proximity (r ~ 0.01) to ensure hydrogeological needs are separately considered. Road and urban proximity were weakly correlated (r = 0.19), reflecting clustered infrastructure and accessibility benefits, while well proximity negatively correlated with drainage density (r = -0.37), highlighting the need to avoid of geomorphologically unstable areas. Overall, no single factor dominated RWH suitability, and the combination of hydrogeological, topographic, and accessibility variables ensured a balanced, physically and operationally meaningful assessment of the RWH suitability potential sites.

Variance Inflation Factor (VIF) analysis was conducted for all the fourteen RWH conditioning factors to assess potential multicollinearity (O'Brien, 2007). All VIF values were well below the common threshold of 5, with slope having the highest value (~2.0) and LULC the lowest (~1.0), indicating minimal collinearity among variables. These results confirm that each factor contributes unique, non-redundant information to the model, supporting reliable feature importance assessment in the Random Forest analysis. The absence of multicollinearity ensures stable parameter estimates and interpretable results, validating the integrity of the final suitability model (O'Brien, 2007). Consequently, all the fourteen factors were retained, maximizing the descriptive and predictive power of the RWH site selection framework (Fig. 6).

3.4. Variable importance factor

Random Forest feature importance analysis revealed a distinct hierarchy in the factors controlling RWH site suitability. Slope emerged as the dominant factor (17.5%), highlighting the critical role of geomorphology in regulating runoff, erosion, and construction feasibility, with gentle slopes favoring water retention. Drainage density (11.5%) and the Topographic Wetness Index (10.1%) further underscored the influence of surface hydrology in identifying areas of natural water accumulation. Socio-economic and accessibility factors well proximity (9.3%), urban proximity (9.1%), and road proximity (8.1%) were secondary yet essential for practical implementation and maintenance. In contrast, climate and subsurface factors, including rainfall (2.2%), land use/land cover (2.4%), and hydrogeology (2.5%), exhibited low predictive power, indicating the primacy of surface conditions in determining suitability (Fig. 7). The relatively low importance of rainfall suggests that, although precipitation is necessary, its spatial variability is less constraining than topography in water capture and retention (Pappaka et al., 2025). The RF model achieved the highest AUC (0.91), indicating strong predictive accuracy and reliability (Kumie, 2019). Overall, RWH site selection should prioritize topographical and hydrological attributes, followed by accessibility considerations, with climatic and subsurface factors serving a supporting role.

The Random Forest model achieved an overall accuracy of 81.4% for RWH site selection, which is generally acceptable in

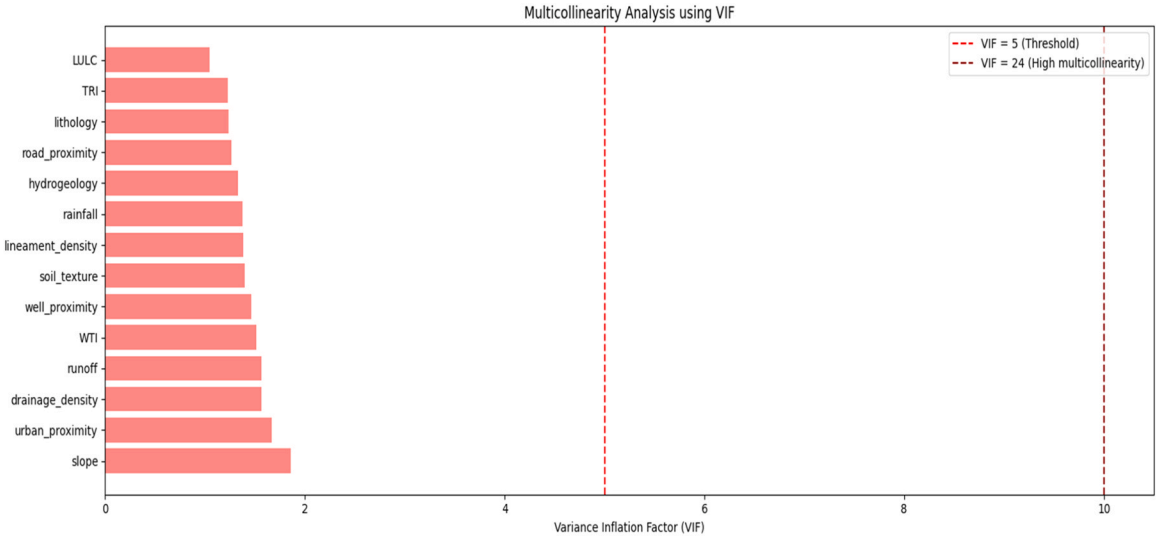


Fig. 6. Multicollinearity Analysis Using VIF.

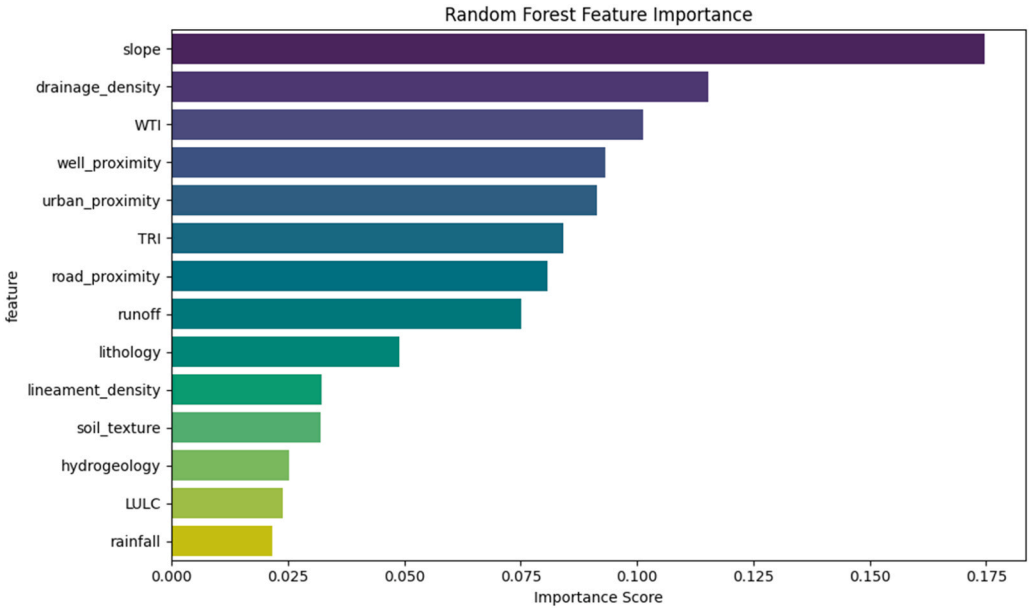


Fig. 7. Relative importance of variables.

geospatial and hydrological modeling contexts (Pappaka et al., 2025). Traditional GIS-based MCDA approaches, such as AHP, often achieve lower accuracies of 50–53%, due to categorical reclassification and subjective weighting (Gebremedhn et al., 2023). Therefore, the 81.4% accuracy exceeds these conventional thresholds, supporting reliable prioritization of sites based on slope, soil, runoff, and proximity factors. Machine learning models, including RF, generally achieve 70–85% accuracy, positioning the current result as reasonable, particularly in data-scarce regions (Pappaka et al., 2025). With proper validation, this performance is sufficient to inform practical RWH site selection and guide targeted improvements in feature selection or ensemble modeling.

3.5. *Results of factors influencing rainwater harvesting suitability*

Suitability analyses for surface water in the UTB reveal distinct spatial patterns across multiple criteria (Fig. 8). Slope-based suitability shows that very low, low, moderate, high, and very high classes occupy 17.6%, 45.3%, 22.5%, 7.7%, and 6.9% of the basin, respectively (Fig. 8J), while soil texture analysis indicates that clay soils (1.3% of the basin) are very highly suitable, with overall infiltration suitability dominated by moderate (66.4%) and high (28.6%) classes (Fig. 8I). LULC strongly influences runoff and infiltration, with vegetated areas promoting infiltration and impervious surfaces enhancing runoff; bare land, grassland, agricultural land, and shrubland (98.4%) are very high to moderate in suitability, whereas waterbodies, settlements, and forest lands (1.6%) show minimal suitability (Fig. 8B). Rainfall-based suitability is mainly low to moderate (16.2% very low, 41.8% low, 30.65% moderate; Fig. 8F), while runoff-based suitability is higher, with moderate (47.3%), high (18.7%), and very high (10.0%) classes collectively covering over three-quarters of the basin (Fig. 8H). Lineament density indicates that areas with no or low density are highly suitable (very high: 72.1%; high: 11.6%), whereas very high-density zones (7.1%) are unsuitable (Fig. 8D). Hydrogeological suitability is dominated by high (87.6%) and very high (10.9%) classes, concentrated along river corridors Fig. 8C, and drainage density highlights very high and high-density zones as most favorable (16.1% and 28.6%), followed by moderate (26.4%), low (18.5%), and very low (10.4%) classes (Fig. 8A). Lithology constrains suitability, with the low class covering 71.2% of the basin (Fig. 8E). Accessibility analyses show that 26.7%, 14.2%, and 11.2% of the basin are very high, high, and moderate suitability based on road proximity, while areas distant from urban centers (very high, high, moderate: 71.6%) minimize contamination and implementation costs (Figs. 8G, 8M). TRI results indicate that very high and high suitability classes occupy 25.0% and 9.1% of the basin, with moderate ruggedness dominating (32.5%) and steep terrains limiting feasibility (Fig. 8K). Well proximity reveals 40.30% of the basin as unsuitable, with high and moderate suitability covering 13.4% and 35.6%, and the most favorable areas (10.7%) concentrated in western, northern, and southeastern sectors (Fig. 8N). WTI indicates limited groundwater potential, with very low and low classes covering 30.3% and 44.0%, moderate 18.9%, and high and very high only 4.6% and 2.2% (Fig. 8L). Collectively, these analyses show that RWH potential is highest in gently undulating, accessible terrains near wells with favorable hydrology, geology, and land cover, whereas steep, rugged, lithological constrained, or low-groundwater areas require careful prioritization.

3.6. *Rainwater harvesting suitability*

From the entire UTB area of 3266 km^2, approximately 607.5 km^2 (18.6%) was identified as potentially suitable for rainwater harvesting (RWH) prior to the application of constraints. These areas were delineated based on slope, soil texture, land use/land cover,

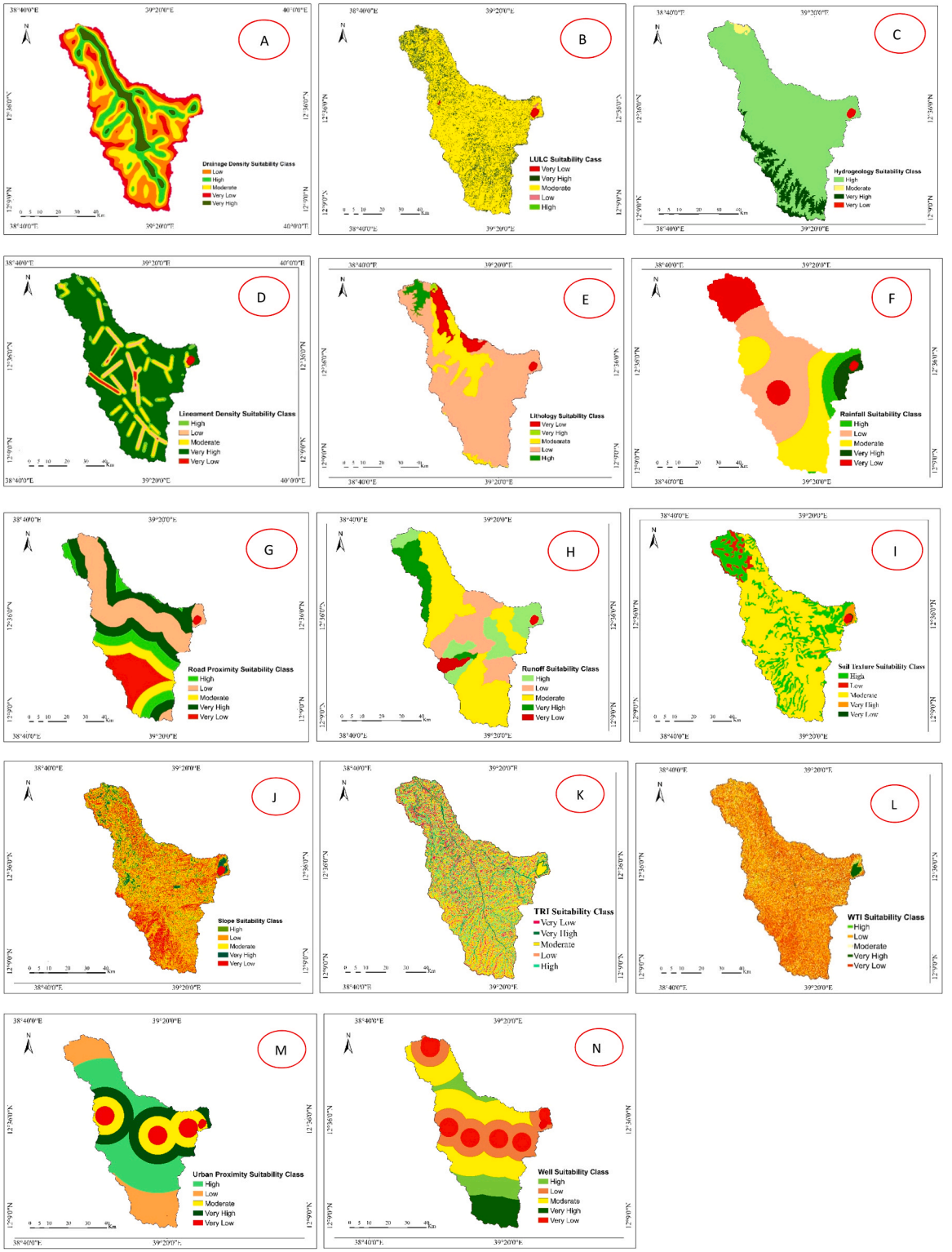


Fig. 8. Sutability map of each criterion based on fuzzy membership functions: a) Drainage Density, b) LULC, c) Hydro geology, d) Lineament Density, e) Lithology, f) Rainfall, g) Road Proximity, h) Runoff, i) Soil Texture, j) Slope, k) TRI, l) WTI, M) Urban proximity, and N) Well Proximity.

proximity to infrastructure, and hydrological indices. Suitable zones were predominantly associated with slopes less than 5%, clay-rich soil textures, bare land and grassland cover, and far distances from wells, roads, and settlements. Hydrologically, these zones coincided with higher drainage density, lower Terrain Ruggedness Index (TRI), higher Topographic Wetness Index (TWI), and relatively higher rainfall and modeled surface runoff. Favorable lithological units and low lineament density were also observed in these areas.

Constraint analysis (Fig. 9) excluded a total of 146.9 km^2, corresponding to 4.6% of the basin area, from RWH consideration. Among the applied constraints, lineaments accounted for the largest excluded area (93.8 km^2; 63.8%), followed by major roads (35.0 km^2; 23.8%) and lakes (18.2 km^2; 12.4%). After constraint implementation, approximately 95.4% of the basin area remained available for RWH suitability classification.

Following the application of constraints, the final RWH suitability map showed that very high suitability covered 74.5 km^2 (2.3%) of the basin, high suitability covered 463.1 km^2 (14.5%), and moderate suitability accounted for 1897 km^2 (59.2%). Low suitability and very low suitable areas covered 317.8 km^2 (9.9%) and 393.7 km^2 (9.48%), respectively (Fig. 9).

The application of constraints resulted in reductions across all suitability classes, amounting to 0.9% for very low, 1.1% for low, 0.8% for moderate, 1.5% for high, and 0.4% for very high suitability classes. Overall, approximately 1.8–1.9% of land initially classified as suitable for RWH was excluded following constraint implementation. The combined high and very high suitability classes decreased from 18.7% to 16.8% of the total basin area.

3.7. Validation of potential RWH

To check the applicability of the choice methodology in the study area for constructing different types of RWH need to be validated with existing RWH technologies for their suitability for RWH sites. For validation, 23 ground-truth sites (11 farm ponds and 12 micro-dams) were used based on the criteria in Table 4. Within the 3266 km^2 study area, 37.41% was identified as suitable for farm ponds, predominantly in the central watershed where grassland and cultivated land dominate. The validation results (Fig. 9) show that 51.14% of existing RWH sites fall within highly suitable zones, while only 14.3% of micro-dams are located in moderately suitable areas, confirming the reliability of the suitability model. Therefore, accuracy assessment taken into consideration by overlaying the locations of existing RWH ponds and the potential RWH zones which were developed by the model were agreed with the experiences and local knowledge of the farmers. The capacity of existing ponds that fall under moderately suitability has low water holding capacity in a given area and due to this most of the ponds were vulnerable to runoff damage, sediment deposition and high Evaporation. As shown in Fig. 9, the existed ponds fall under very high suitable and high suitable lands, which confirms the present RWH suitability is accurate (Mosase et al., 2017).

A comparison of the RWH suitability map and the location of the existing farm pond shows that the pond is located in the suitable demarcated by the study, which indicates that the methodology adopted has a very good capability to identify locations suitable for RWH.

4. Discussion

The spatial distribution of RWH suitability in the UTB reflects the combined influence of topographic, soil, hydrological, and land-use characteristics. The predominance of moderate suitability (59.2%) indicates substantial spatial heterogeneity across the subbasin, where favorable conditions for RWH are present but may require site-specific engineering or management interventions to enhance effectiveness.

The relatively limited reduction in suitable area following constraint application (approximately 2%) suggests that high-potential RWH zones in the UTB are largely located away from major infrastructural and environmental exclusion zones. This is consistent with the subbasin's rugged terrain, where settlements and road networks are preferentially concentrated in valley bottoms and accessible corridors that are already less favorable for surface runoff-based RWH due to slope or hydrological limitations. Consequently, constraint buffers overlapped more frequently with moderate and low suitability classes than with high or very high suitability zones.

Lineaments accounted for the largest proportion of excluded area, highlighting their extensive spatial distribution and their importance as preferential subsurface flow pathways. Areas with high lineament density may facilitate rapid infiltration and subsurface drainage, reducing the effectiveness of surface-based RWH structures. The dominance of lineament-related exclusions underscores the importance of integrating geological factors into RWH suitability assessments to avoid overestimation of feasible sites.

The observed reduction in high and very high suitability classes from 18.7% to 16.8% aligns with findings reported by Harka et al. (2020), who documented a 5–10% reduction in feasible RWH sites after applying exclusion criteria such as road buffers and protected areas. The comparatively smaller reduction in the present study suggests that hydrologically favorable RWH locations in the UTB are inherently less constrained by anthropogenic and environmental exclusion zones.

Comparisons with other Ethiopian studies further support these findings. In the Waghmare zone of northern Ethiopia, Alene et al. (2022) reported approximately 7% highly suitable, 17.8% moderately suitable, 30.3% less suitable, and 44.9% unsuitable areas for RWH using a GIS-based approach, attributing high suitability to regions with gentle slopes and fine-textured soils. Similarly, Gebremedhn et al. (2023) identified suitable lands for supplemental irrigation in the Ethiopian highlands using GIS-MCDA, emphasizing slope and runoff potential, although higher overall suitability (approximately 50–60%) was reported due to less rugged terrain compared to the UTB.

In broader semi-arid contexts, Chen et al. (2025) reported that approximately 12% of arid regions exhibited high suitability for RWH using integrated modeling approaches, which is comparable to the combined high and very high suitability identified in the UTB. Studies from Indian drylands have reported even higher proportions of high-to-moderate suitability (>70%), reflecting differences in

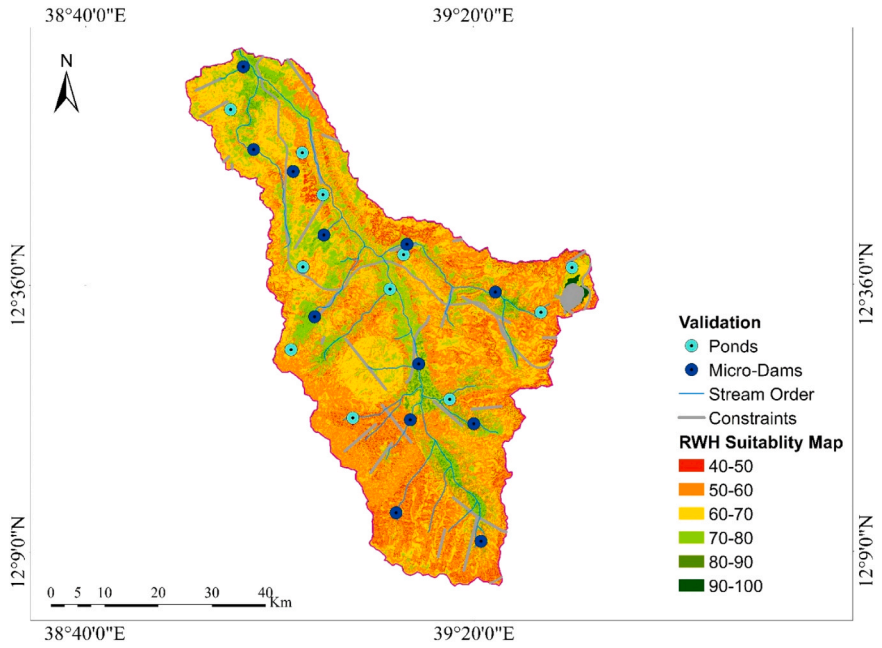


Fig. 9. Rainwater harvesting (RWH) suitability map of the UTB, with validation points from existing RWH structure.

topography, soil depth, and rainfall regimes.

The extensive moderate suitability zones identified in the UTB represent a key opportunity for expanding RWH through targeted engineering measures such as channel lining, sediment traps, and small-scale storage structures. Overall, the finding that approximately 16.8% of the UTB is practically suitable for RWH after constraint implementation is consistent with basin-scale assessments in the Ethiopian Highlands, including Demisachew (2018), who reported final feasible RWH areas ranging from 10% to 35% after incorporating technical and socio-economic constraints. These results demonstrate that incorporating constraints enhances the realism and applicability of RWH suitability mapping for watershed scale water resources planning.

5. Conclusions and recommendations

5.1. Conclusions

The Upper Tekeze Basin (UTB) is a semi-arid system where high rainfall variability and complex topography constrain water availability and limit effective runoff utilization. This study develops an integrated, basin-scale framework for rainwater harvesting (RWH) suitability assessment by coupling SWAT-based hydrological modeling with Random Forest and geospatial analysis.

The results show that RWH potential is primarily controlled by topography and surface hydrology, particularly slope, drainage density, and topographic wetness index. Incorporation of practical constraints refines feasible implementation zones, reducing the combined high and very high suitability classes from 18.6% to 16.78% of the basin area. Field verification indicates strong agreement between high-suitability zones and existing RWH structures, supporting the reliability of the approach.

From a regional perspective, the suitability maps provide a screening and prioritization tool for identifying areas where RWH interventions are most effective. However, the analysis is limited to basin-scale assessment; site-specific storage design, temporal runoff variability, and detailed water balance components were not explicitly addressed. Future work should integrate hydrological outputs with engineering design and volumetric analysis at selected sites. Overall, the framework provides a transferable and hydrologically consistent basis for supporting decentralized water resource planning and climate adaptation in semi-arid regions.

5.2. Recommendations

Areas classified as high and very high suitability should be prioritized for immediate RWH implementation, while extensive moderate suitability zones can be improved through cost-effective measures such as sediment control, channel stabilization, and small-scale storage enhancement. RWH structures should be preferentially located in areas with low lineament density and less permeable lithological conditions to minimize subsurface losses.

For effective application, the suitability outputs should be translated into decision-support tools for basin authorities and local planners, supported by targeted capacity building and post-implementation monitoring. The proposed framework is transferable and can be applied to other data-scarce semi-arid basins to support hydrologically informed and climate-resilient water resource management.

Future research should focus on integrating climate change scenarios (e.g., RCP4.5 and RCP8.5) to assess the long-term performance of RWH systems, and on developing nature-based solutions to reduce seepage losses in high-permeability zones.

CRediT authorship contribution statement

Tewodros T. Assefa: Writing – review & editing, Supervision. **Negash T. Roba:** Writing – review & editing, Visualization, Validation, Investigation, Conceptualization. **Daniel W. Mebrie:** Formal analysis, Data curation, Conceptualization. **Abadefar Demelash Debebe:** Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Ethical approval

The study was retrospective; therefore, formal consent was not required.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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